

Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

```
clc;
```

```
clear;
```

```
close all;
```

```
%% --- Input Parameters ---
```

```
Fs = 10000;    % Sampling frequency (Hz)
```

```
Fc = 2500;     % High-pass cutoff frequency (Hz)
```

```
N = 50;       % Filter order
```

```
% Normalized cutoff frequency
```

```
wc = Fc/(Fs/2);
```

```
%% --- FIR High-Pass Filter using Hamming Window ---
```

```
b_hamm = fir1(N, wc, 'high', hamming(N+1));
```

```
%% --- Display Filter Coefficients ---
```

```
disp('--- FIR High-Pass Filter Coefficients (Hamming Window) ---');
```

```
disp(b_hamm');
```

```
%% --- Frequency Response ---
```

```
figure;
```

```
freqz(b_hamm, 1, 1024, Fs);
```

```
title('FIR High-Pass Filter using Hamming Window');
```

```
%% --- Magnitude Response ---
```

```
[H, f] = freqz(b_hamm, 1, 1024, Fs);
```

```
figure;
```

```
plot(f, 20*log10(abs(H)));
```

```
grid on;
```

```
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude (dB)');
```

```
title('Magnitude Response of FIR High-Pass Filter');
```

```
%% --- Phase Response ---
```

```
figure;
```

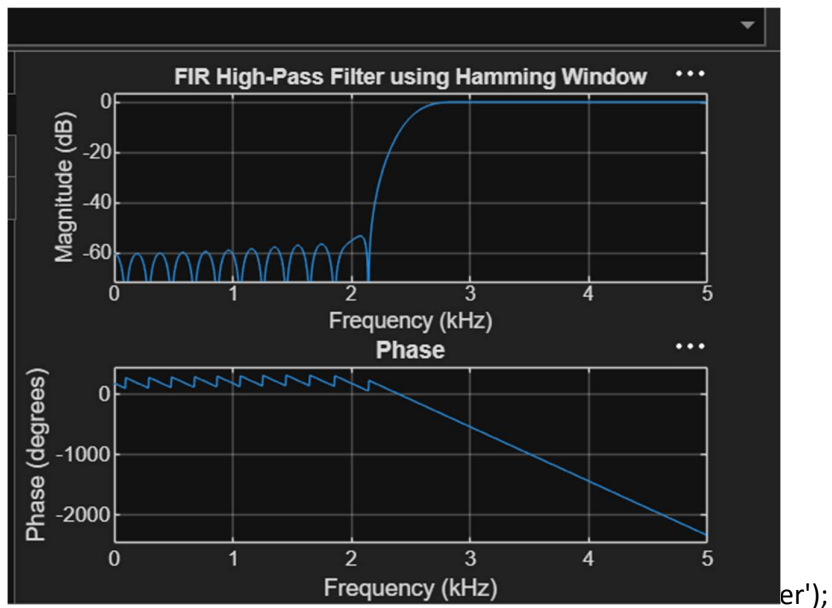
```
plot(f, unwrap(angle(H)));
```

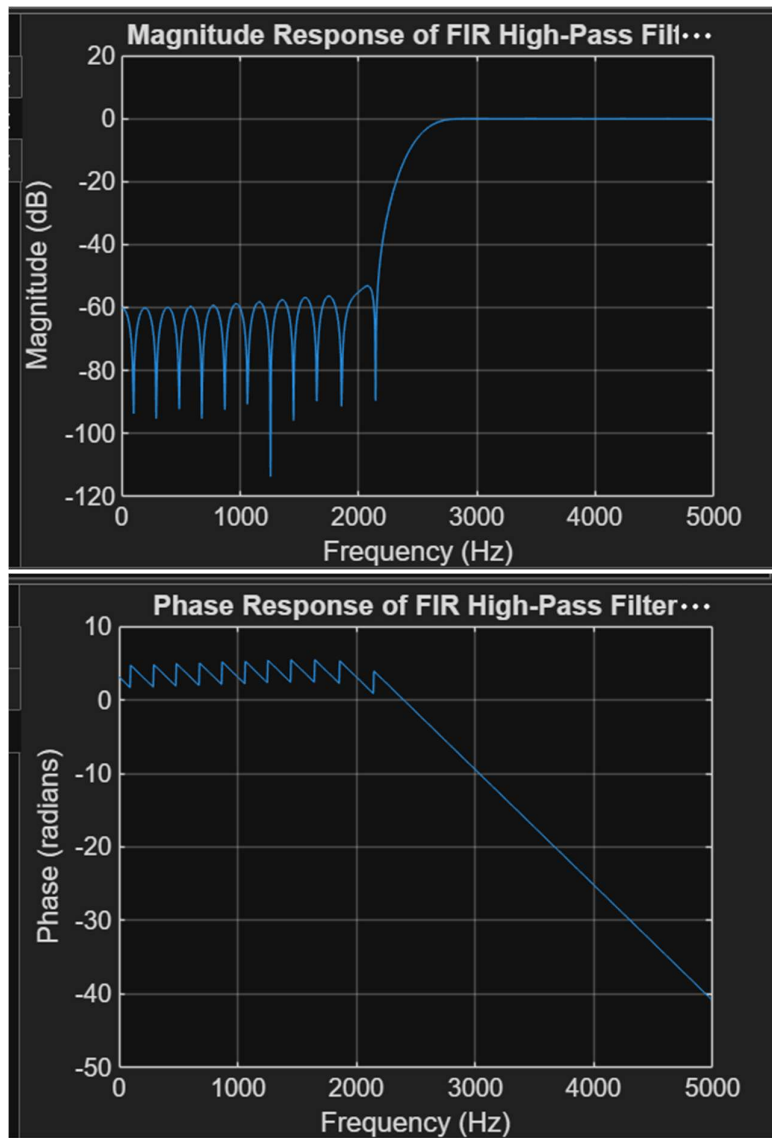
```
grid on;
```

```
xlabel('Frequency (Hz)');
```

```
ylabel('Phase (radians)');
```

```
title('Phase Response of FIR High-Pass Filt
```





Result: 1. The FIR High-Pass Filter was successfully designed using the Hamming Window technique for the specified cutoff frequency, sampling frequency, and filter order.

2. The FIR filter coefficients were successfully generated using the Hamming Window method in MATLAB and the filter was implemented.

3. The magnitude response and phase response of the designed FIR High-Pass Filter were obtained and analyzed. The magnitude response showed effective attenuation of low-frequency components and passage of high-frequency components, confirming the desired high-pass filtering behavior. The phase response exhibited the expected linear-phase characteristics of an FIR filter.